Contextualised Competency Mapping (CCM)

Competency Requirements of the Position

POSITION ANALYSED	systems architect
POSITION DESCRIPTION	systems architect
COMPANY	systems architect
CCM ASSESSMENT DATE	02 September 2025
ASSESSORS	Elmien Toerien

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Introduction

The CCM

The Contextualised Competency Mapping (CCM) technique is an online questionnaire-based job analysis system which allows job experts (the assessor) to clarify the work-related competency requirements of a position/job/role.

The CCM not only assesses job competency requirements, but the performance and psychological characteristics of a person or team can be linked to the role requirements.

A number of CCM reports are available for each position analysed, including:

- Competency Requirements of the Position
- Cognitive Person–Role Matching Report
- Cognitive Team–Role Matching Report
- 360 Degree Person-Role Matching Report
- Organisational Mapping Report, the Cognitive Diversity Heatmap & the Cultural Diversity Heatmap

About this report

The **Competency Requirements of the Position** reflects the views of job experts who assessed the job-related cognitive and competency requirements of the role.

The various CCM reports can, as part of a holistic assessment battery, be used for the purposes of:

- the clarification of work-related competency requirements
- the anchoring and contextualisation of competency assessment results
- talent auditing
- job and organisational structuring
- selection and placement

Description of assessor

Assessor

The following assessor evaluated the SST / RO work environment and the cognitive competency requirements of the position.

Name of Assessor	Toerien, E
Job Title	Other
Company	Cognadev IT
Relation to position	Colleague Manager

Disclaimer:
This report was generated electronically by the Cognadev CCM Expert System. The use of the results is limited to accredited practitioners who have received specialist training by Cognadev in the use and application of the CCM. Due consideration should be given to the subjective nature of the interpretation and application of the results. Cognadev accepts no liability, of any kind, for the consequences of the use of this report.

Item analysis

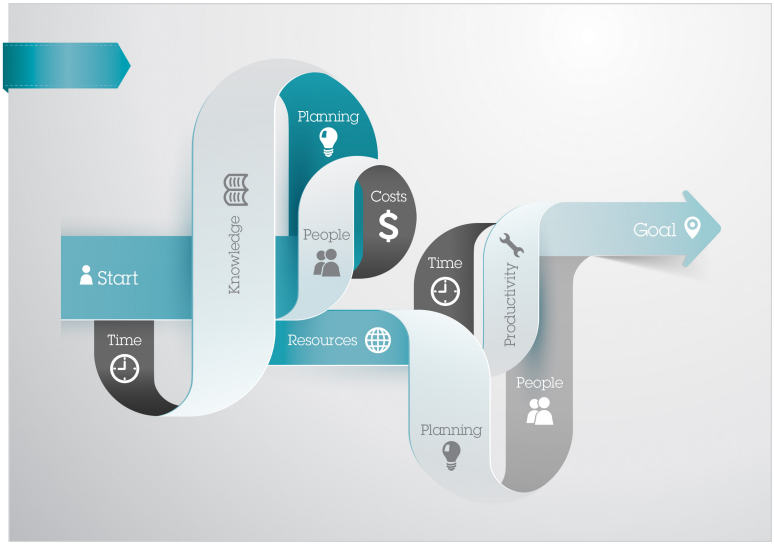
The following table indicates the cumulative weights or percentage of scores that were allocated by the assessor to the items reflecting various levels of work complexity, namely the Pure Operational, Diagnostic Accumulation, Tactical Strategy, Parallel Processing and Pure Strategic environments. The highest score allocations per item are highlighted in blue.

Section	Items	Distribution of scores				
		Operational	Diagnostic	Tactical	Parallel	Strategic
A Generic description of work	1. Nature of work	-	-	100	-	-
	2. Focus	-	-	100	-	-
	3. Role	-	-	100	-	-
	4. Responsibility	-	-	-	100	-
	5. Goals	-	-	-	100	-
	6. Time frame	-	-	100	-	-
B Specific work-related functions	1. Performance criteria	-	-	100	-	-
	2. Subject matter	-	-	-	100	-
	3. Contents dealt with	-	-	-	-	100
	4. Knowledge and experience	-	-	-	100	-
	5. Language	-	-	-	100	-
	6. Management	-	-	100	-	-
	7. Problems	-	-	100	-	-
	8. Guidance	-	-	100	-	-
	9. Decision making	-	-	-	100	-
	10. Actions	-	-	-	100	-
	11. Innovation	-	-	-	100	-
	12. Personal orientation	-	-	100	-	-
C Processing requirements	1. Exploration	-	-	100	-	-
	2. Analysis	-	-	100	-	-
	3. Structuring	-	-	-	100	-
	4. Integration	-	-	-	100	-
	5. Logical reasoning	-	-	-	100	-
	6. Conceptualisation and verbalisation	-	-	100	-	-
	7. Memory	-	-	100	-	-
	8. Self-awareness	-	-	-	100	-
	9. Learning	-	-	-	-	100

Current complexity of work requirements of the position

Based on the weights allocated by the assessor to the various SST / RO levels of work, the systems architect role largely involves the following:

Tactical Strategy / Alternative Paths work



- Optimising systems efficiencies
- Formulate functional strategy
- Benchmarking
- Goal achievement of a functional unit
- Create alternative strategic paths
- Evaluate and implement systems
- Planning and resource allocation
- Follow theoretical guidelines
- Project management
- Professional work

A Tactical Strategy / Alternative Paths work environment can be described as follows:

Type of work

- Works with a whole operating system or functional unit

Ways of working

- Optimises the goal achievement of the functional unit using various tactics
- Uses specialised knowledge of operational work and incorporates theory and experience
- Finds best practice solutions to problems
- Makes the best use of all available resources (human, material and financial)
- Evaluates systems and practices and implements new systems
- Professional practice

Information used

- Complex, concrete and abstract
- Meaningful wholes, systems, plans, budgets and networks
- Ambiguous goals and outcomes
- Relatively stable contexts

Outputs / achievements

- Ambiguous and are judged in terms of operating efficiencies
- Cannot be predicted linearly but must be extrapolated from the principles of systems functioning

Judgement

- Relates to planning and implementation to optimise resources

Language used

- Focuses on the symbolic (consumables, equipment and resources)

Learning

- Hypothesising, implementation and continuous evaluation of theoretical and practical knowledge

Time frame

- Ranges from one year to three years for the most complex tasks

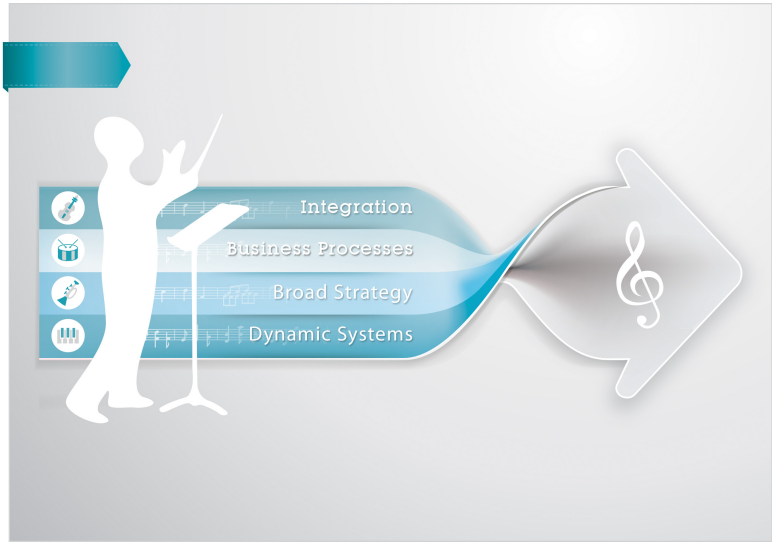
Examples

- First level of organisational improvement
- Management of operating systems (divisional, middle, senior or general manager)
- Professionals and specialists operating independently

Potential complexity of work requirements of the position

Based on the weights allocated by the assessors to the various SST / RO levels of work, the systems architect role largely involves the following:

Parallel Processing work



- Integrate dynamic systems (value chains, supply-demand factors)
- Synchronise parallel pathways
- Use trade-offs to maintain progress
- Coordinate functional units
- Formulate business processes and broad business strategy
- Ensure organisational viability
- Systems focused
- Maintain the pace of and monitor interacting projects

A Parallel Processing work environment can be described as follows:

Type of work

- Focuses on business processes and business strategy
- Involves co-ordinating the work of various functional or business units
- Ensures that the organisation remains viable

Ways of working

- Plans and implements long-term business solutions
- Translates abstract ideas and decisions into strategies and tactics
- Plans how the organisation is to be managed to achieve its mission statement
- Manages and co-ordinates several interacting projects, or programmes simultaneously in order to achieve an overall objective
- Manoeuvres resources between interacting projects to capitalise on economies of scale
- Builds models
- Has true accountability

Information used

- Primarily complex, unfamiliar and ambiguous
- Often abstract, theoretical and generalised, such as long-term policies, broad strategies, values, business processes and solutions
- Ambiguous realities and goals
- Unstable and unpredictable contexts

Outputs / achievements

- Often abstract and theoretical
- Organisational change and transformation
- Development of new functionalities and models

Judgement

- Profit and loss accountability, directing the implementation of strategies to ensure organisational viability

Language used

- A conceptual focus (talent pool, culture, values, capital accumulation, assets, labour)

Learning

- Hypothesising and testing abstract conceptual models and strategies

Time frame

- Ranges from three years to five years for the most complex task

Examples

- General management, executive and CEO levels
- Chief specialist roles in the organisation
- Developers of new functionalities

Notes: On the generated level of work complexity of the position as calculated by the CCM report.

Should the position clearly reflect a single level of work complexity, only one work environment is reported on here. However, the position may reflect both a Current and a Potential level of work complexity.

Processing competencies

Given the complexity requirements of the Tactical Strategy / Alternative Paths work environment, the following cognitive competencies can be expected to optimise job functioning. The top four processing skills as identified by the assessor are reported on first, followed by the rest in no particular order. It should be pointed out that although cognitive capability may be a prerequisite for optimal job functioning, many other factors may also be involved.

Tactical Strategy / Alternative Paths competencies

The top four ranked cognitive competency requirements of the position include:

Processing Competency	Description in terms of the systems architect role
Learning	• learning to optimally respond to situations and opportunities • rapid identification of operational patterns and proactive responses to that • curiosity and application of an explorative and learning approach
Exploration	• active investigation of functional and operational aspects • curiosity about alternative approaches and methods • unbiased, goal directed searches in a relatively unstructured environment • hypothesise to attach meaning to seemingly unrelated occurrences • discriminate between relevant and irrelevant elements given their pragmatic and operational implications
Analysis	• differentiate between subcomponents of approaches and methodologies • link alternative options • identify relationships between operational procedures and their contribution to goal achievement • detailed and precise analysis of budgets, business plans, professional issues and operational procedures • accuracy in managing systems
Structuring	• focus on meaningful wholes and structures (such as models, tangible / defined systems, organising schemes) • order, structure and generalise information • categorise, classify, taxonomise, represent and map information • extract core information rather than focusing on peripheral system elements • apply metacognitive strategies to cope with detail complexity

Additional processing skills required by the position:

Processing Competency	Description in terms of the systems architect role
Conceptualisation	• clear and coherent verbalisation of ideas • communicate operational strategies and tactics • explain system functioning
Logical reasoning	• reason about the practical implications of innovative operational procedures • reason about the viability of theoretical approaches • devise 1- 3 year business plans • justifying resource allocation decisions with logical evidence • operationalise abstract structure and function decisions in terms of concrete actions
Memory	• well-established knowledge base and memory of specialist technical information • capitalise on previous experience in complex tangible contexts • tendency to automate core elements of recurring events for pattern identification and extrapolation
Metacognition	• awareness of own biases and assumptions • the formulation of thinking strategies to improve operations • the use of particular criteria to evaluate products and procedures • evaluate own plans and decisions regarding operational functions • awareness of own thinking processes and managerial style • planning change and discontinuation according to criteria for goal achievement

Depending on the role the following processing characteristics may also be important to optimise job functioning:

Processing Tendencies	Description in terms of Tactical Strategy / Alternative Paths
General Response style	• pace control yet readiness to act • active rather than passive approach • rapid, timely implementation of insights • avoid impulsive conclusions and quick closure • quick insight at an intuitive level • apply a well-structured approach
Flexibility	• formulate alternative approaches and methodologies via lateral thinking • redefine functionality • openness to alternatives available in the market • embrace technological innovation and complexity • curiosity and explorative approach and general open-mindedness

Job-specific competencies

Behavioural competencies involve a combination of knowledge, experience and psychological factors. These involve a combination of knowledge, experience and psychological factors that are associated with effective work performance. The identification of job-related competencies is useful for purposes of anchoring various HR functions aimed to drive and support business success such as selection, placement, performance appraisal, development, succession and promotion. It is particularly useful in describing role requirements and in guiding individual performance expectations.

The job assessor(s) have selected the following 8 to 12 most relevant job-related competencies required for optimal job performance in the role of the systems architect position.

Core Competency requirements for the systems architect position:

Confidence	Understanding oneself, believing in capabilities and intuition. Having willpower and intentions to achieve purpose and direction. Experiencing a sense of personal substance.
Learning orientation	Curiosity, flexibility and self-awareness to acquire new skills, coping strategies and understanding through investigation, experimentation, practice and the integration of feedback.
Inspirational Leadership	Required skill, confidence and insight to exert social influence to accomplish a common goal by providing guidance, direction, information, inspiration and acting as a role model.
Planned and structured	The tendency to order, describe and represent information in a systematic manner. This includes the observation of appropriate rules and to adequately plan an approach.
Technical knowledge	An understanding or capacity to learn a specific knowledge base, plus gain experience in its application. This requires previous or future educational, theoretical and practical exposure.
Customer service orientation	Ensuring quality service by understanding the needs, requirements and cultural orientation of customers. Building customer relationships and being responsive to customer feedback..
Quality and standards	Awareness of certain criteria to ensure a high degree of value / excellence of products and processes. Also, the ability to implement high standards and to not compromise.
Communication	The capacity to effectively interact with others. Listening skills to interpret and convey information in a clear, interesting, accurate and appropriate manner to a particular audience.
Negotiation and conflict resolution	Skill in objectively directing and participating in dialogue between parties intended to reach an understanding or agreement to resolve points of difference or competing interests. Reframing conflicting perspectives to facilitate a compromise or resolution.
Computer literacy	The capacity to use computers effectively. This includes the use of operational systems, networks, hardware, programming, advanced problem-solving and architecting skills.
Visionary and strategic	Creation of a clear direction based on the consideration of various ideas and opportunities. Forward thinking, pro-active and broad-based perspective ensuring the long-term viability of the organisation..

Knowledge and Skill

The following areas of job-related knowledge and skills were identified by the job assessor as crucial for optimal job performance. The required degree of expertise was also rated on a 5-point scale where a 1 indicates superficial knowledge and / or skill and a 5 indicates in-depth technical understanding and / or skill in application.

Areas / Disciplines	Job requirements (knowledge, skill or both)	Degree required
Communications	Both	4
Hospitality	Skills	3
Entertainment	Skills	5

Derailers

Inadequate functioning on any of these performance areas may inhibit / derail effectiveness in the position of systems architect.

Derailers identified in the position of systems architect.
Experience and skills
Cognition: technical / analytical orientation
Cognition: integrative, holistic orientation

Final comments

The CCM is a job analysis system aimed at identifying work-related competency requirements. It was developed and is distributed by Cognadev.

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